

Def Agent- benchmarks C.)

(#4)

⇒ Consider tasks that require tool use (e.g., running code) and iterating over a period of time

⇒ Agent includes: LLM, memorization, tool, workflow

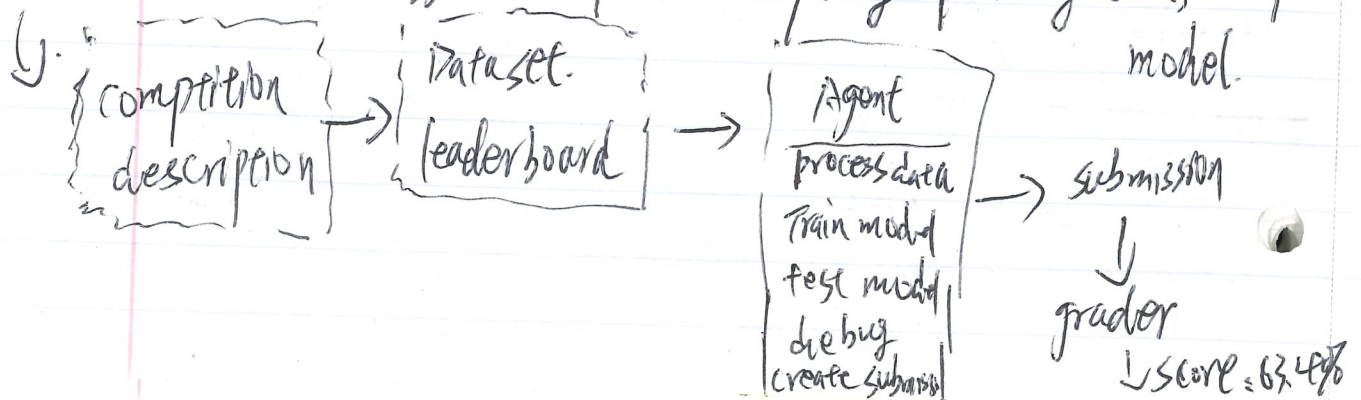
2). Agent = language model + agent scaffolding

SWEBench: { 2294 tasks across 12 python repos: (logic for deciding how to use the LLM)
Given codebase + issue description
Evaluation metric: Unit test ⇒ Submit a PR.

paper: SWEBench: Can language Models Resolve Real-world Github issues? - Jimenez, et al., - 2024

CyBench: A Framework for Evaluating Cybersecurity capabilities and Risks of language models - Zhang + 2024
includes 40 capture the Flag (CTF) tasks
use first-resolve time as a measure of difficulty

KALE Bench: 75 Kaggle competitions: requiring processing data, train/test model.



Def pure-reasoning-benchmarks(): (#5)

All tasks require linguistic and world knowledge
Can we isolate reasoning from knowledge?

Arguably, reasoning captures a more pure form of intelligence
(isn't just about memorizing facts)

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ARC-AGI-1

ARC-AGI-2: François Chollet: creator of Keras lib.

(Abstract-Reasoning corpus - Artificial General Intelligence)

AGI: is not only a system that can automate most of economically valuable work,

but also a system that can automatically acquire new skills / knowledges outside of its training data

safety

~~safe~~ ~~safety~~

Def ~~safe~~ safety-benchmarks(): (#6)

what does safety mean for AI?

HAI

(HELM safety: curated set of benchmarks): ^{HAI}stanford website

↓ Harm/Bench: Based on 510 harmful behaviors that violate laws

↓ ATR-Bench: Based on regulatory frameworks

or concerns

↓ Jailbreaking:

Taxonomized into 314 risk categories and company policies

⇒ LLM Refuses harmful instructions after 5694 prompts

⇒ Greedy coordinate Gradient automatically optimizes prompts to bypass safety

⇒ Transfer from open-weight models (Llama) to closed models (GPT-4)